AQUÍ VA EL TÍTULO DEL ARTÍCULO

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ABSTRACT. Aquí va el resumen

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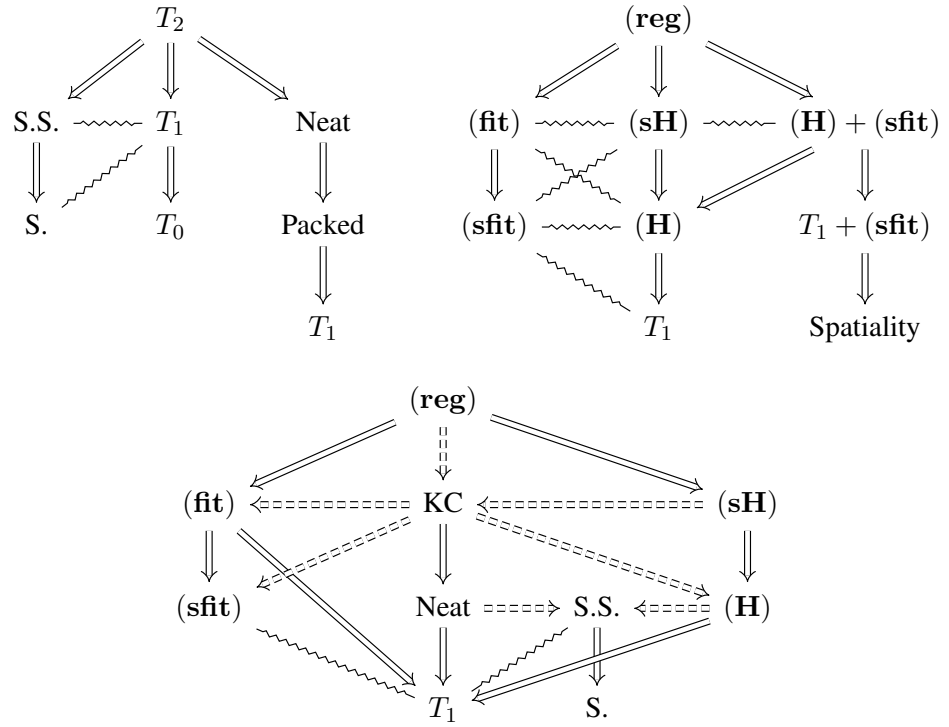
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1. INTRODUCTION

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2. PRELIMINARIES

2.1. **Frames, nuclei, and open filters.** We recall the basic notions and notation concerning frames, nuclei, and open filters; for general background, see the standard references on point-free topology.

A *frame* is a complete lattice A satisfying

$$a \wedge \bigvee X = \bigvee \{a \wedge x \mid x \in X\}.$$

A frame homomorphism preserves finite meets and arbitrary joins. We write $\mathbf{Frm}$ for the category of frames. The Heyting implication on A is characterized by

$$x \wedge a \leq b \quad \Longleftrightarrow \quad x \leq (a \succ b).$$

Every frame homomorphism $f : A \to B$ has a right adjoint $f_* : B \to A$, characterized by $f(a) \leq b$ if and only if $a \leq f_*(b)$.

A *nucleus* on A is a map $j : A \to A$ satisfying

$$x \leq j(x), \qquad j(j(x)) = j(x), \qquad j(x \wedge y) = j(x) \wedge j(y).$$

The fixed-point set $A_j = \{x \in A \mid j(x) = x\}$ is a frame, with joins given by $\bigvee_{A_j} X = j(\bigvee_A X)$. The map $j^* : A \to A_j$, $j^*(x) = j(x)$, is a surjective frame homomorphism, and every quotient arises in this way. We denote by NA the frame of nuclei on A. For $a \in A$, put

$$u_a(x) = a \vee x, \qquad v_a(x) = a \succ x, \qquad w_a(x) = (x \succ a) \succ a.$$

The nuclei u_a and v_a determine the closed and open quotients, respectively.

An element $p \in A$ is $\wedge$-*irreducible* if $p \neq 1$ and $x \wedge y \leq p$ implies $x \leq p$ or $y \leq p$. The set $\mathrm{pt}(A)$ of such elements is endowed with the topology generated by

$$\mathcal{U}(a) = \{p \in \mathrm{pt}(A) \mid a \not\leq p\}, \qquad a \in A.$$

The assignment $\mathcal{U} : A \to \mathcal{O}(\mathrm{pt}(A))$ is a surjective frame homomorphism, called the *spatial reflection*. A frame is *spatial* when $\mathcal{U}$ is an isomorphism. Points correspond bijectively to completely prime filters through $F_p = \{a \in A \mid a \not\leq p\}$.

A filter F on A is *open* if, for every directed $D \subseteq A$,

$$\bigvee D \in F \quad \Longrightarrow \quad D \cap F \neq \varnothing.$$

We write $A^\wedge$ for the collection of open filters. If $f : A \to B$ is a frame homomorphism with right adjoint f_*, then

$$G \in B^\wedge \Longrightarrow f_*G \in A^\wedge, \qquad f_*G = \{a \in A \mid f(a) \in G\}. \tag{1}$$

Proposition 2.1. *Let $f = f^* \colon A \to B$ be a frame morphism. If $G \in B^\wedge$, then $f_*G \in A^\wedge$.*

Every nucleus j determines its *admissible filter*

$$\nabla(j) = \{a \in A \mid j(a) = 1\}.$$

Given a filter F, the least nucleus admitting F is

$$v_F = f_F^\infty = \bigvee_{a \in F} v_a, \qquad f_F(x) = \bigvee_{a \in F} (a \succ x), \tag{2}$$

where the join defining v_F is taken in NA. For $F \in A^\wedge$, one has $\nabla(v_F) = F$. Thus, open filters correspond bijectively to fitted nuclei whose admissible filters are open.

The Hofmann–Mislove theorem identifies $A^\wedge$ with the compact saturated subsets of $\mathrm{pt}(A)$. Explicitly, the corresponding sets and filters are

$$Q_F = \{p \in \mathrm{pt}(A) \mid (\forall a \in F)\ a \not\leq p\}, \qquad F_Q = \{a \in A \mid Q \subseteq \mathcal{U}(a)\}.$$

We refer to (F, Q_F) as an ***HM pair***. When $\mathrm{pt}(A) = \varnothing$, the only open filter is A, corresponding to $Q_A = \varnothing$ and to the trivial fitted quotient $v_A = \top_{NA}$.

2.2. **Compact quotients and admissibility intervals.** Recall that the quotient A_j determined by a nucleus j is compact if and only if $\nabla(j)$ is an open filter. In particular, for each $F \in A^\wedge$, the fitted nucleus v_F determines a compact quotient. However, a compact quotient need not be uniquely determined by its admissible filter.

For $F \in A^\wedge$, let

$$M_F = \{m \in A \setminus F \mid m \text{ is maximal in } A \setminus F\}.$$

Define

$$w_F(x) = \bigwedge \{m \in M_F \mid x \leq m\}$$

and the nucleus w_F is the greatest nucleus with admissible filter F, whereas v_F is the least one. Consequently,

$$\{j \in NA \mid \nabla(j) = F\} = [v_F, w_F].$$

We call this the *admissibility interval* associated with F. Evaluating at 0 yields

$$[v_F(0), w_F(0)] \subseteq A.$$

If $\mathrm{pt}(A)$ is T_1, then $M_F = Q_F$, identifying points with their corresponding $\wedge$-irreducible elements.

We shall also use nuclei induced by subspaces. For a topological space S and $E \subseteq S$, define

$$[E] : \mathcal{O}S \longrightarrow \mathcal{O}S, \qquad [E](U) = (E \cup U)^\circ.$$

A nucleus of this form is called *spatially induced*.

Let Q be a compact saturated subset of S, put $G = \{U \in \mathcal{O}S \mid Q \subseteq U\}$, and write $Q' = S \setminus Q$. Then

$$\nabla([Q']) = G, \qquad v_G \leq [Q'] \leq w_G.$$

More generally, let $F \in A^\wedge$, let $Q = Q_F$, and put $G = F_Q = \mathcal{V}(Q)$, viewed as an open filter of $\mathcal{O}(\mathrm{pt}(A))$. We call (F, Q, G) an *HM triple*. Denote by

$$\mathcal{U}_* : \mathcal{O}(\mathrm{pt}(A)) \to A$$

the right adjoint of the spatial reflection.

Proposition 2.2. *For every $k \in [v_G, w_G]$, the composite*

$$\mathcal{U}_* \circ k \circ \mathcal{U}$$

is a nucleus on A with admissible filter F.

Hence Proposition 2.5 gives the map

$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

and, in particular,

$$v_F \leq \Phi(k) \leq w_F.$$

Finally, when $A = \mathcal{O}S$, identifying S with $\mathrm{pt}(A)$, the spatial reflection is an isomorphism.

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Lemma 2.3. *If $A = \mathcal{O}S$, then $\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E]$ for every $E \subseteq S$.*

3. Separation properties induced by compact quotients

The notions studied in this section are motivated by the work of Sexton and Simmons on point-sensitive and point-free patch constructions. Our aim is to identify nuclei associated with open filters that determine compact closed quotients, and to investigate the separation properties arising from this requirement. We begin by recalling Sexton's notion of neatness ([6, Definition 4.3] and [8, Definition 8.2.1]), which provides the starting point for our approach.

Definition 3.1. Let A be a frame, let $F \in A^\wedge$ and $\alpha \in \mathrm{Ord}$. We say that:

- F is α-neat, if

$$x \vee d = 1, \quad \text{for every } x \in F,$$

  where $d = f_F^{\alpha}(0)$.
- A is α-neat if every $F \in A^{\wedge}$ is α-neat.
- A is neat if it is α-neat for some α.

Let us record a useful consequence of neatness. Suppose that A is α-neat and let $F \in A^{\wedge}$. Put $d = f_F^{\alpha}(0)$. Since $x \vee d = 1$ for every $x \in F$, we have $F \subseteq \nabla(u_d)$. Thus, $f_F \leq u_d$, and consequently

$$v_F = f_F^{\infty} \leq u_d.$$

On the other hand,

$$d = f_F^{\alpha}(0) \leq v_F(0),$$

so that

$$u_d \leq u_{v_F(0)} \leq v_F.$$

Therefore,

$$v_F = f_F^{\infty} = u_d.$$

In particular, every fitted nucleus associated with an open filter is closed, and hence every such compact quotient is simultaneously closed.

Sexton showed that a topological space is neat if and only if it is both packed and stacked [6, Theorem 6.2]. Motivated by this spatial characterization, we now introduce point-free counterparts of stackedness and strong stackedness.

Let S be a topological space, let $Q \subseteq S$ be compact saturated, and let

$$G = \mathcal{V}(Q) \in (\mathcal{O}S)^{\wedge}.$$

Recall that S is stacked if

$$v_G(\varnothing) = (Q')^{\circ}$$

for every such pair (G, Q), and strongly stacked if

$$v_G = [Q']$$

for every such pair.

We now return to the general setting. Let A be a frame, $S = \mathrm{pt}(A)$, and let (F, Q, G) be an H–M triple. Recall that the map of Proposition 2.2

$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

provides a natural way of comparing the admissibility interval associated with G with that associated with F.

Proposition 3.2. *Let (F, Q, G) be an H–M triple. Then $\Phi(w_G) = w_F$, that is,*

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

Proof. By Proposition 2.2, we have

$$\Phi(w_G) = U_* \circ w_G \circ U \leq w_F.$$

For the reverse inequality, by adjunction it suffices to show that

$$U \circ w_F \leq w_G \circ U.$$

Since $w_G = [M']$, for every $x \in A$ we have

$$w_G(U(x)) = [M'](U(x)) = (M' \cup U(x))^\circ.$$

Thus, it is enough to prove that

$$U(w_F(x)) \subseteq M' \cup U(x).$$

Suppose that $y \in U(w_F(x))$ and $y \notin M'$. Since $y \notin M'$, we have $y \in M$. Moreover, $y \in U(w_F(x))$ implies $w_F(x) \not\leq y$. But

$$w_F(x) = \bigwedge\{m \in M \mid x \leq m\}.$$

Hence $x \not\leq y$, since otherwise $w_F(x) \leq y$, a contradiction. Therefore $y \in U(x)$, proving the desired inclusion. Consequently, $\Phi(w_G) = w_F$. □

Proposition 3.2 shows that Φ preserves the upper elements of the corresponding admissibility intervals. Since $[Q']$ is the spatially induced nucleus appearing in Sexton's definitions, the nucleus

$$\Phi([Q']) = U_* \circ [Q'] \circ U$$

provides a natural point-free counterpart. This motivates the following definitions.

Definition 3.3. Let A be a frame and (F, Q) an H–M pair. We say that:

- A is *FQ-stacked* if
$$(U_* \circ [Q'] \circ U)(0) = v_F(0).$$
- A is *FQ-strongly stacked* if
$$U_* \circ [Q'] \circ U = v_F.$$

The frame A is *stacked* (respectively, *strongly stacked*) if it is FQ-stacked (respectively, FQ-strongly stacked) for every H–M pair (F, Q).

Clearly, every strongly stacked frame is stacked. Let us look that these properties are conservative. We first establish the relation between the transfinite iterations defining the fitted nuclei on a space and on its frame of open sets.

Lemma 3.4. *Let $\alpha \in \mathrm{Ord}$ and $X \in CS$. Then*

$$\partial_F^\alpha(X) = \big(f_F^\alpha(X')\big)'.$$

In particular, for every $U \in \mathcal{O}S$,

$$\big(\partial_F^\infty(U')\big)' = v_F(U).$$

Proof. We proceed by transfinite induction on α. For $\alpha = 0$, the identity follows from $\partial_F^0 = \mathrm{id} = f_F^0$.

Suppose that the identity holds for α. Then

$$\begin{aligned}\partial_F^{\alpha+1}(X) &= \partial_F(\partial_F^\alpha(X)) \\ &= \big(f_F((\partial_F^\alpha(X))')\big)' \\ &= \big(f_F(f_F^\alpha(X'))\big)' \\ &= \big(f_F^{\alpha+1}(X')\big)'.\end{aligned}$$

If λ is a limit ordinal, then

$$\begin{aligned}\partial_F^\lambda(X) &= \bigcap_{\alpha<\lambda} \partial_F^\alpha(X) \\ &= \bigcap_{\alpha<\lambda} (f_F^\alpha(X'))' \\ &= \left(\bigcup_{\alpha<\lambda} f_F^\alpha(X')\right)' \\ &= \big(f_F^\lambda(X')\big)'.\end{aligned}$$

Thus, the identity holds for every ordinal α. Taking the stable value of the iterations yields the last assertion. □

Proposition 3.5. *The notions of stackedness and strong stackedness are conservative.*

Proof. Let S be a topological space and put $A = \mathcal{O}S$. For every H–M pair (F, Q), we have $v_F = [Q']$ whenever S is strongly stacked. By Lemma 2.3,

$$U_* \circ [Q'] \circ U = [Q'] = v_F.$$

Thus, A is strongly stacked.

Conversely, if A is strongly stacked, then

$$v_F = U_* \circ [Q'] \circ U = [Q']$$

for every H–M pair (F, Q), and hence S is strongly stacked.

Now suppose that S is stacked. By definition,

$$Q = \partial_F^\infty(S).$$

Consequently,

$$Q' = (\partial_F^\infty(S))' = v_F(\varnothing),$$

where the last equality follows from Lemma 3.4. By Lemma 2.3,

$$\begin{aligned}(U_* \circ [Q'] \circ U)(\varnothing) &= [Q'](\varnothing) \\ &= (Q')^\circ \\ &= Q' \\ &= v_F(\varnothing).\end{aligned}$$

Therefore, A is stacked.

Conversely, if A is stacked, then

$$v_F(\varnothing) = (U_* \circ [Q'] \circ U)(\varnothing) = Q'.$$

By Lemma 3.4,

$$(\partial_F^\infty(S))' = Q',$$

and hence $\partial_F^\infty(S) = Q$. Thus, S is stacked. □

Neatness requires every fitted nucleus associated with an open filter to determine a closed compact quotient. A natural extension is to impose this condition on every nucleus admitting an open filter. This leads to a point-free counterpart of Wilansky's classical KC property [11].

Definition 3.6. Let A be a frame. We say that A is *KC* if every compact quotient of A is closed.

In the spatial setting, Wilansky's KC property coincides with Sexton's notion of packedness. Consequently,

$$\text{Neat} \iff \text{KC and Stacked}.$$

Whether analogous relationships hold in the point-free setting will be addressed in Section 4. We first examine how these properties interact with classical separation axioms.

3.1. **Comparison with Classical Separation Axioms.** Although stackedness and strong stackedness are conservative, neither property is comparable with the T_1 separation axiom, as the following examples show.

Example 3.7.

(1) Consider the frame $A = [0, 1]$ with its usual order. Its open filters are

$$F_r = (r, 1], \qquad 0 \le r < 1,$$

and the corresponding compact saturated subsets of $\operatorname{pt} A = [0, 1)$ are $Q_r = [0, r]$. A direct computation gives

$$v_{F_r}(x) = \begin{cases} x, & x \le r, \\ 1, & x > r, \end{cases} = (U_* \circ [Q_r'] \circ U)(x).$$

Thus, A is strongly stacked but not T_1.

(2) The Hausdorff frame of Paseka and Šmarda [3], considered in Example **??**, is T_1 but not stacked, and hence not strongly stacked.

Recall that every T_1 stacked space is sober and that, in the strongly stacked case, $w_G = v_G$. We now investigate analogous phenomena for frames.

Proposition 3.8. *Let A be a frame. If*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then every point of A is maximal in $\operatorname{pt} A$.

Proof. Let $p, q \in \operatorname{pt} A$ with $p \leq q$, and consider the open filter

$$F_q = \{x \in A \mid x \not\leq q\}.$$

By the definition of w_{F_q},

$$w_{F_q} = v_q, \qquad \text{so} \qquad w_{F_q}(0) = q.$$

By hypothesis,

$$q = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x.$$

For every $x \in F_q$, we have $x \not\leq q$, and hence $x \not\leq p$. Since p is a point,

$$x \wedge \neg x = 0 \leq p \quad \Longrightarrow \quad \neg x \leq p.$$

Consequently,

$$q = \bigvee_{x \in F_q} \neg x \leq p.$$

Thus $p = q$, proving that every point of A is maximal. □

Proposition 3.9. *Let A be a T_1 frame. Then:*

(1) *A is stacked if and only if*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$.

(2) *A is strongly stacked if and only if*

$$v_F = w_F$$

for every $F \in A^\wedge$.

Proof. Let (F, Q) be an H–M pair. Since A is T_1, we have $Q = M$, and hence $w_G = [Q']$. By Proposition 3.2,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F.$$

Both assertions now follow directly from the definitions of stackedness and strong stackedness. □

The preceding characterizations assume that the frame is T_1. We now consider conditions under which the corresponding identities imply the T_1 separation property.

Corollary 3.10. *Let A be a frame such that*

$$\mathcal{V}(a) = \{p \in \operatorname{pt} A \mid a \leq p\} \neq \varnothing$$

for every $a \neq 1$. If

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then A is T_1. In particular, the same conclusion holds if $v_F = w_F$ for every $F \in A^\wedge$.

Proof. By Proposition 3.8, every point of A is maximal in $\mathrm{pt}\,A$. Suppose that there exist $p \in \mathrm{pt}\,A$ and $a \in A$ such that $p < a < 1$. By hypothesis, there is $q \in \mathrm{pt}\,A$ with $a \leq q$. Hence

$$p < a \leq q,$$

contradicting the maximality of p in $\mathrm{pt}\,A$. Therefore, every point of A is maximal in A, and hence A is T_1. The last assertion follows immediately from $v_F = w_F \Rightarrow v_F(0) = w_F(0)$. □

3.2. **Inheritance and preservation properties.** The notions of stacked and strongly stacked frames introduced above are formulated in terms of the admissibility intervals associated with Hofmann–Mislove pairs. A closely related phenomenon had already appeared in the work of Simmons.

More precisely, Simmons shows in [9, Lemma 8.9 and Corollary 8.10] that, for a Hofmann–Mislove pair (F, Q), with $G = \nabla(Q)$, the diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ {\scriptstyle \mathcal{U}}\downarrow & & \downarrow{\scriptstyle \mathcal{U}} \\ \mathcal{O}S & \xrightarrow[v_G]{} & \mathcal{O}S \end{array} \tag{3}$$

commutes laxly; that is,

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Here

$$v_F = f_F^{\infty} \qquad \text{and} \qquad v_G = f_G^{\infty}$$

are the fitted nuclei associated with $F \in A^{\wedge}$ and $G \in (\mathcal{O}S)^{\wedge}$, respectively.

In the terminology developed in the preceding sections, Simmons' inequality may be viewed as an early manifestation of the behavior underlying our notions of stackedness: it provides a canonical comparison between the lower bounds of the corresponding admissibility intervals in A and $\mathcal{O}S$.

We now show that this phenomenon is not restricted to the spatial reflection. In fact, an analogous comparison holds for an arbitrary frame morphism. Let

$$h \colon A \longrightarrow B$$

be a frame morphism, let $G \in B^{\wedge}$, and set

$$F = h_* G \in A^{\wedge}.$$

We shall compare the fitted nuclei determined by these filters through the diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h} \\ B & \xrightarrow[v_G]{} & B. \end{array}$$

As in Simmons' situation, the relevant comparison is first established at the level of the prenuclei whose transfinite iterations generate the corresponding fitted nuclei. The following lemma provides the basic inequality needed for this purpose.

Lemma 3.11. *Let $h\colon A \to B$ be a frame morphism, $G \in B^\wedge$, and $F = h_*G$. Then*

$$h \circ f_F \leq f_G \circ h.$$

Proof. Recall that, by (1),

$$F = h_*G = \{a \in A \mid h(a) \in G\}.$$

Hence

$$f_F = \bigvee\{v_a \mid a \in F\} \qquad \text{and} \qquad f_G = \bigvee\{v_y \mid y \in G\}.$$

Let $x \in A$. For every $a \in F$, we have $h(a) \in G$, and therefore

$$h(v_a(x)) = h(a \to x) \leq h(a) \to h(x) = v_{h(a)}(h(x)) \leq f_G(h(x)).$$

Thus, since h preserves arbitrary joins,

$$\begin{aligned} h(f_F(x)) &= h\left(\bigvee_{a\in F} v_a(x)\right) \\ &= \bigvee_{a\in F} h(v_a(x)) \\ &\leq f_G(h(x)). \end{aligned}$$

Hence

$$h \circ f_F \leq f_G \circ h.$$

□

The preceding inequality is preserved throughout the transfinite iteration of the corresponding prenuclei.

Lemma 3.12. *Let $h\colon A \to B$ be a frame morphism, $G \in B^\wedge$, and $F = h_*G$. Then, for every ordinal α,*

$$h \circ f_F^\alpha \leq f_G^\alpha \circ h.$$

In particular,

$$h \circ v_F \leq v_G \circ h.$$

Proof. We proceed by transfinite induction on α. If $\alpha = 0$, the result is immediate.

For the successor step, suppose that the inequality holds for α. Then, by Lemma 3.11 and the induction hypothesis,

$$\begin{aligned} h \circ f_F^{\alpha+1} &= h \circ f_F \circ f_F^\alpha \\ &\leq f_G \circ h \circ f_F^\alpha \\ &\leq f_G \circ f_G^\alpha \circ h \\ &= f_G^{\alpha+1} \circ h. \end{aligned}$$

Finally, let λ be a limit ordinal. For every $a \in A$, since h preserves arbitrary joins, we have

$$\begin{aligned} h\big(f_F^\lambda(a)\big) &= h\left(\bigvee_{\alpha<\lambda} f_F^\alpha(a)\right) \\ &= \bigvee_{\alpha<\lambda} h\big(f_F^\alpha(a)\big) \\ &\leq \bigvee_{\alpha<\lambda} f_G^\alpha(h(a)) \\ &= f_G^\lambda(h(a)), \end{aligned}$$

where the inequality follows from the induction hypothesis. Hence

$$h \circ f_F^\lambda \leq f_G^\lambda \circ h.$$

Since

$$v_F = f_F^\infty \qquad \text{and} \qquad v_G = f_G^\infty,$$

taking an ordinal at which both transfinite iterations have stabilized yields

$$h \circ v_F \leq v_G \circ h.$$

This completes the proof. □

Proposition 3.13. *If A is a neat frame, then A_j is a neat frame.*

Proof. Suppose that A is α-neat. Let $G \in A_j^\wedge$ and set

$$F = j_* G \in A^\wedge.$$

Consider the corresponding prenuclei

$$f_F = \bigvee_{x \in F} v_x \qquad \text{and} \qquad f_G = \bigvee_{y \in G} v_y.$$

Since A is α-neat, for every $x \in F$,

$$x \vee \widehat{d} = 1, \qquad \text{where } \widehat{d} = f_F^\alpha(0).$$

Set

$$d = f_G^\alpha(0).$$

By Proposition 3.12, applied to the frame morphism $j = j^* \colon A \to A_j$,

$$j(\widehat{d}) = j\big(f_F^\alpha(0)\big) \leq f_G^\alpha(j(0)) = f_G^\alpha(0) = d.$$

Since j is inflationary,

$$\widehat{d} \leq j(\widehat{d}) \leq d.$$

Now let $y \in G$. Since $y \in A_j$, we have $j(y) = y$. Thus, by (1),

$$y \in F = j_* G.$$

Consequently,

$$y \vee \widehat{d} = 1.$$

Since $\widehat{d} \leq d$, it follows that

$$1 = y \vee \widehat{d} \leq y \vee d \leq 1.$$

Hence

$$y \vee d = 1.$$

Therefore, A_j is α-neat, and hence neat. □

The following proposition establishes that neatness is preserved under binary coproducts.

Proposition 3.14. *Let A_1 and A_2 be α-neat frames. Then $A_1 \oplus A_2$ is α-neat.*

Proof. Let

$$\iota_i \colon A_i \longrightarrow A_1 \oplus A_2, \qquad i = 1, 2,$$

be the canonical frame morphisms, and let

$$F \in (A_1 \oplus A_2)^{\wedge}.$$

For $i = 1, 2$, set

$$F_i = (\iota_i)_* F \in A_i^{\wedge}.$$

Since A_i is α-neat, if

$$d_i = f^{\alpha}_{F_i}(0),$$

then

$$x_i \vee d_i = 1$$

for every $x_i \in F_i$.

By the definition of $F_i = (\iota_i)_* F$,

$$x_i \in F_i \quad \Longleftrightarrow \quad \iota_i(x_i) \in F.$$

Under the usual representation of the binary coproduct, this gives

$$x_1 \oplus 1, \ 1 \oplus x_2 \in F.$$

Since F is a filter,

$$(x_1 \oplus 1) \wedge (1 \oplus x_2) = x_1 \oplus x_2 \in F.$$

Now set

$$d = f^{\alpha}_{F}(0 \oplus 0).$$

By Proposition 3.12, applied to the canonical morphisms ι_i, we have

$$\iota_i \circ f^{\alpha}_{F_i} \leq f^{\alpha}_{F} \circ \iota_i, \qquad i = 1, 2.$$

Evaluating at $0 \in A_i$, we obtain

$$\iota_1(d_1) \leq f^{\alpha}_{F}(0 \oplus 1) \qquad \text{and} \qquad \iota_2(d_2) \leq f^{\alpha}_{F}(1 \oplus 0).$$

Therefore,

$$\begin{aligned} d_1 \oplus d_2 &= (d_1 \oplus 1) \wedge (1 \oplus d_2) \\ &= \iota_1(d_1) \wedge \iota_2(d_2) \\ &\leq f_F^{\alpha}(0 \oplus 1) \wedge f_F^{\alpha}(1 \oplus 0) \\ &= f_F^{\alpha}\big((0 \oplus 1) \wedge (1 \oplus 0)\big) \\ &= f_F^{\alpha}(0 \oplus 0) \\ &= d. \end{aligned}$$

Finally, for $x_i \in F_i$,

$$\begin{aligned} 1 \oplus 1 &= (x_1 \vee d_1) \oplus (x_2 \vee d_2) \\ &= (x_1 \oplus x_2) \vee (d_1 \oplus d_2) \\ &\leq (x_1 \oplus x_2) \vee d \\ &\leq 1 \oplus 1. \end{aligned}$$

Hence

$$(x_1 \oplus x_2) \vee d = 1 \oplus 1.$$

Thus $A_1 \oplus A_2$ is α-neat. □

Corollary 3.15. *Let A_1 and A_2 be α-neat and β-neat, respectively. Then $A_1 \oplus A_2$ is γ-neat, where*

$$\gamma = \max\{\alpha, \beta\}.$$

Proof. Let $\gamma = \max\{\alpha, \beta\}$. Then A_1 and A_2 are both γ-neat. Hence, by Proposition 3.14, $A_1 \oplus A_2$ is γ-neat. □

We extend the previous result to arbitrary coproducts.

Theorem 3.16. *Neatness is preserved under arbitrary coproducts.*

Proof. Let $\{A_i\}_{i \in I}$ be a family of neat frames, and for each $i \in I$, let α_i be a neatness degree of A_i. Consider

$$A = \bigoplus_{i \in I} A_i$$

and let

$$\iota_i \colon A_i \longrightarrow A$$

denote the canonical coproduct morphisms.

Let $F \in A^{\wedge}$. For each $i \in I$, set

$$F_i = (\iota_i)_* F \in A_i^{\wedge}.$$

Moreover,

$$x_i \in F_i \quad \Longleftrightarrow \quad \iota_i(x_i) \in F.$$

Set

$$\alpha = \sup\{\alpha_i \mid i \in I\}.$$

Then every A_i is α-neat. Hence, for every $x_i \in F_i$,

$$x_i \vee d_i = 1,$$

where

$$d_i = f_{F_i}^{\alpha}(0).$$

Let

$$f_F = \bigvee_{x \in F} v_x$$

be the prenucleus associated with F. By Proposition 3.12, applied to each canonical coproduct morphism $\iota_i \colon A_i \to A$, we have

$$\iota_i \circ f_{F_i}^{\alpha} \leq f_F^{\alpha} \circ \iota_i.$$

Evaluating at $0 \in A_i$, we obtain

$$\iota_i(d_i) = \iota_i\big(f_{F_i}^{\alpha}(0)\big) \leq f_F^{\alpha}(\iota_i(0)) = f_F^{\alpha}(\langle 0 \rangle).$$

Set

$$d = f_F^{\alpha}(\langle 0 \rangle).$$

Therefore,

$$\iota_i(d_i) \leq d$$

for every $i \in I$. Taking arbitrary joins yields

$$\langle d_i \rangle \vee \bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let

$$\langle x_i \rangle \in F.$$

Since $x_i \vee d_i = 1$ for every $i \in I$, we have

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle.$$

As $\langle d_i \rangle \leq d$,

$$\langle 1 \rangle = \langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d \leq \langle 1 \rangle.$$

Thus,

$$\langle x_i \rangle \vee d = \langle 1 \rangle.$$

Therefore,

$$A = \bigoplus_{i \in I} A_i$$

is α-neat, and hence neat. □

The preceding results have a categorical consequence. Indeed, the subcategory of neat frames is closed under quotients, by Proposition 3.13, and under arbitrary coproducts, by Proposition 3.16. Hence, the standard criterion for coreflectivity of full subcategories of Frm applies.

Corollary 3.17. *The subcategory of neat frames is a coreflective subcategory of* Frm.

We now turn to the inheritance of the stronger properties introduced above. We first show that strong stackedness is inherited by quotients, and then establish the analogous result for the KC property.

Recall that, for a frame morphism $f\colon A \to B$, $j \in NA$, and $k \in NB$, we have

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NB$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where f_* denotes the right adjoint of f.

Proposition 3.18. *Let A be a strongly stacked frame and $j \in NA$. Then A_j is strongly stacked.*

Proof. Let $F \in A_j^{\wedge}$, and set

$$S = \operatorname{pt} A, \qquad S_j = \operatorname{pt} A_j.$$

Consider the open filter

$$G = j_*(F) = \{x \in A \mid j(x) \in F\},$$

and let

$$Q = S_j \setminus F, \qquad R = S \setminus G.$$

Denote by

$$\mathcal{U}\colon A \longrightarrow \mathcal{O}S \qquad \text{and} \qquad \mathcal{U}_j\colon A_j \longrightarrow \mathcal{O}S_j$$

the corresponding spatial reflections.

Set

$$l = N(\mathcal{U}_j)_*([Q']) = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j$$

and

$$m = N(j^*)(v_G).$$

By Proposition 2.2,

$$v_F \leq l.$$

Thus, it is enough to prove

$$l \leq m \leq v_F.$$

We first prove that $m \leq v_F$. Since $G = j_*(F)$, the fitted nuclei associated with G and F satisfy

$$j^* \circ v_G \leq v_F \circ j^*. \tag{4}$$

Moreover,

$$m = N(j^*)(v_G) = \bigvee_{x \in A} \left(u_{j^*(v_G(x))} \wedge v_{j^*(x)}\right).$$

By (4), for every $x \in A$,

$$j^*(v_G(x)) \leq v_F(j^*(x)),$$

and hence

$$u_{j^*(v_G(x))} \leq u_{v_F(j^*(x))}.$$

Therefore,

$$u_{j^*(v_G(x))} \wedge v_{j^*(x)} \leq u_{v_F(j^*(x))} \wedge v_{j^*(x)} \leq v_F.$$

Taking joins over $x \in A$, we obtain

$$m \leq v_F. \tag{5}$$

We now prove that $l \leq m$. Since A is strongly stacked,

$$v_G = N(\mathcal{U})_*([R']) = \mathcal{U}_* \circ [R'] \circ \mathcal{U}.$$

Hence,

$$m = N(j^*)\big(N(\mathcal{U})_*([R'])\big).$$

To compare the u-components in the decompositions of l and m, consider the auxiliary nucleus

$$n = N(\mathcal{U}_j \circ j^*)_*([Q']).$$

We first observe that, for every $x \in A_j$,

$$j^*(n(x)) = l(x).$$

Indeed,

$$\begin{aligned} j^*(n(x)) &= j^* \left(j_* \circ (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j \circ j^*\right)(x) \\ &= j^* j_* \left(\bigvee \left\{a \in A_j \;\middle|\; \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j^*(x))\right\}\right) \\ &= \bigvee \left\{a \in A_j \;\middle|\; \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(x)\right\} \\ &= l(x). \end{aligned}$$

Now, under the usual identification of S_j with the corresponding subspace of S, we have

$$Q = R \cap S_j.$$

Using the decomposition of a nucleus in terms of its u- and v-components, we have

$$l = \bigvee_{x \in A_j} \left(u_{l(x)} \wedge v_x\right) = \bigvee_{x \in A_j} \left(u_{j^*(n(x))} \wedge v_x\right).$$

On the other hand, since

$$m = N(j^*)\big(N(\mathcal{U})_*([R'])\big),$$

we have

$$m = \bigvee_{a \in A} \left(u_{j^*(N(\mathcal{U})_*([R'])(a))} \wedge v_{j^*(a)}\right).$$

For $x \in A_j$, the explicit description of the right adjoint gives

$$\begin{aligned} j^*\big(N(\mathcal{U})_*([R'])(x)\big) &= j^* \left(\bigvee \left\{a \in A \;\middle|\; \mathcal{U}(a) \subseteq R' \cup \mathcal{U}(x)\right\}\right) \\ &= \bigvee \left\{j^*(a) \in A_j \;\middle|\; \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\right\}. \end{aligned}$$

Moreover, since $x \in A_j$,

$$\mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)$$

is equivalent, after restriction to S_j, to

$$\mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(x).$$

Consequently,

$$j^*(n(x)) \leq j^*\big(N(\mathcal{U})_*([R'])(x)\big).$$

Therefore,

$$u_{j^*(n(x))} \leq u_{j^*(N(\mathcal{U})_*([R'])(x))}$$

for every $x \in A_j$, and hence

$$u_{l(x)} \wedge v_x = u_{j^*(n(x))} \wedge v_x \leq u_{j^*(N(\mathcal{U})_*([R'])(x))} \wedge v_x \leq m.$$

Taking joins over $x \in A_j$, we obtain

$$l \leq m.$$

Combining this inequality with (5) and $v_F \leq l$, we obtain

$$v_F \leq l \leq m \leq v_F.$$

Hence,

$$v_F = l = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j.$$

Therefore, A_j is strongly stacked. □

In particular, since every strongly stacked frame is stacked, we also obtain that A_j is stacked.

We next show that the KC property is also inherited by quotients.

Proposition 3.19. *Let A be a KC frame and $j \in NA$. Then A_j is KC.*

Proof. Let $k \in N(A_j)$ be such that $(A_j)_k$ is compact. Then

$$\nabla(k) \in A_j^{\wedge}.$$

By Proposition 2.1,

$$j_*(\nabla(k)) \in A^{\wedge}.$$

Consider the nucleus

$$l = j_* \circ k \circ j^* \in NA.$$

Since

$$\nabla(l) = j_*(\nabla(k)),$$

the quotient A_l is compact. Since A is KC, there exists $a \in A$ such that

$$l = u_a.$$

We claim that

$$k = u_{j^*(a)}.$$

Indeed, from $l = u_a$ we have

$$a \vee x = l(x) = j_*\big(k(j^*(x))\big)$$

for every $x \in A$. Applying j^* to both sides yields

$$j^*(a) \vee j^*(x) = k(j^*(x)),$$

since $j^* \circ j_* = \mathrm{id}_{A_j}$.

Now let $x \in A_j$. Since $j^*\colon A \to A_j$ is surjective, we have $j^*(x) = x$, and therefore

$$\begin{aligned} u_{j^*(a)}(x) &= j^*(a) \vee x \\ &= j^*(a) \vee j^*(x) \\ &= k(j^*(x)) \\ &= k(x). \end{aligned}$$

Hence

$$k = u_{j^*(a)}.$$

Thus $(A_j)_k$ is closed. Therefore, A_j is KC. □

We conclude this section by examining more closely the relation between the compact quotients of a frame and those arising from its quotients. Besides the inheritance results established above, the comparison obtained in Corollary 3.12 induces a natural frame morphism between the corresponding compact quotients.

By Proposition 3.12, $j \circ v_F \leq v_G \circ j$. We now compare the compact quotients induced by F and G. Consider

$$A_F = \{x \in A \mid v_F(x) = x\} \qquad \text{and} \qquad A_{j,G} = \{y \in A_j \mid v_G(y) = y\}.$$

Define

$$H\colon A \longrightarrow A_{j,G}, \qquad H = v_G \circ j.$$

Since v_F is inflationary,

$$v_G \circ j \leq v_G \circ j \circ v_F.$$

On the other hand, by Proposition 3.12 and the idempotency of v_G,

$$v_G \circ j \circ v_F \leq v_G \circ v_G \circ j = v_G \circ j.$$

Hence

$$H \circ v_F = H.$$

Therefore, H factors through the quotient A_F, inducing a map

$$h\colon A_F \longrightarrow A_{j,G}$$

such that

$$h \circ v_F = v_G \circ j.$$

Explicitly,

$$h(x) = v_G(j(x)), \qquad x \in A_F.$$

Lemma 3.20. *The map*

$$h\colon A_F \longrightarrow A_{j,G}$$

preserves arbitrary joins.

Proof. Let $X \subseteq A_F$. Since joins in A_F are given by

$$\bigvee_{A_F} X = v_F\left(\bigvee_{A} X\right),$$

we have

$$\begin{aligned} h\left(\bigvee_{A_F} X\right) &= v_G\left(j\left(v_F\left(\bigvee_{A} X\right)\right)\right) \\ &\leq v_G\left(v_G\left(j\left(\bigvee_{A} X\right)\right)\right) \\ &= v_G\left(j\left(\bigvee_{A} X\right)\right), \end{aligned}$$

where the inequality follows from Proposition 3.12 and the equality from the idempotency of v_G.

Since

$$H = v_G \circ j \colon A \longrightarrow A_{j,G}$$

is a frame morphism,

$$v_G\left(j\left(\bigvee_{A} X\right)\right) = \bigvee_{A_{j,G}} \{H(x) \mid x \in X\} = \bigvee_{A_{j,G}} h[X].$$

Therefore,

$$h\left(\bigvee_{A_F} X\right) \leq \bigvee_{A_{j,G}} h[X].$$

Conversely, for every $x \in X$,

$$x \leq \bigvee_{A_F} X,$$

and hence, by monotonicity of h,

$$h(x) \leq h\left(\bigvee_{A_F} X\right).$$

Since this holds for every $x \in X$,

$$\bigvee_{A_{j,G}} h[X] \leq h\left(\bigvee_{A_F} X\right).$$

Thus,

$$h\left(\bigvee_{A_F} X\right) = \bigvee_{A_{j,G}} h[X].$$

Therefore, h preserves arbitrary joins. □

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

Proposition 3.21. *The diagram*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle h} \\ A_j & \xrightarrow[v_G]{} & A_{j,G} \end{array} \tag{6}$$

is commutative.

Proof. By Lemma 3.20, h is a frame morphism. Moreover, by the definition of h,

$$h \circ v_F = v_G \circ j.$$

Hence the diagram commutes. □

We now specialize the construction obtained above to the spatial reflection. Let A be a frame and set $S = \operatorname{pt} A$. For $F \in A^\wedge$, let $Q \in \mathcal{Q}S$ be the compact saturated subset associated with F by the Hofmann–Mislove correspondence, and put $G = \nabla(Q) \in (\mathcal{O}S)^\wedge$.

Taking $j = \mathcal{U}_A \colon A \longrightarrow \mathcal{O}S$ in diagram (6), we obtain the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle h} \\ \mathcal{O}S & \xrightarrow[v_G]{} & (\mathcal{O}S)_G. \end{array}$$

By [8, Lemma 5.7.1], the point space of the compact quotient A_F is precisely Q; that is,

$$\operatorname{pt}(A_F) = Q.$$

Hence, the spatial reflection of A_F is

$$\mathcal{U}_{A_F} \colon A_F \longrightarrow \mathcal{O} \operatorname{pt}(A_F) = \mathcal{O}Q.$$

On the other hand, since $G = \nabla(Q)$, the points of the compact quotient $(\mathcal{O}S)_G$ are precisely the points of Q. Thus,

$$\operatorname{pt}((\mathcal{O}S)_G) = Q,$$

and its spatial reflection is

$$\mathcal{U}_{(\mathcal{O}S)_G} \colon (\mathcal{O}S)_G \longrightarrow \mathcal{O}Q.$$

Consequently, the preceding constructions fit into the following commutative diagram:

$$\begin{array}{ccccc} A & \xrightarrow{v_F} & A_F & & \\ {\scriptstyle \mathcal{U}_A}\downarrow & & {\scriptstyle h}\downarrow & \searrow^{\mathcal{U}_{A_F}} & \\ & & & & \mathcal{O}Q \\ & & & \nearrow_{\mathcal{U}_{\mathcal{O}S_\nabla}} & \\ \mathcal{O}S & \xrightarrow[v_\nabla]{} & \mathcal{O}S_\nabla & & \end{array} \tag{7}$$

Since this diagram is determined by $F \in A^\wedge$ and its associated compact saturated subset $Q \in \mathcal{Q}S$, we shall refer to it as the Q-square.

Although property $(\mathbf{H})$ does not in general guarantee that compact quotients are closed, the particular compact quotient appearing in (7) exhibits a closely related behavior.

Theorem 3.22. *Under the hypotheses of the Q-square, if A satisfies property $(\mathbf{H})$, then*

$$\mathcal{O}Q \cong\uparrow Q'.$$

Equivalently, the frame of open sets of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies $(\mathbf{H})$. Since $(\mathbf{H})$ is conservative, $S = \operatorname{pt} A$ is a T_2 space. Hence, for every $Q \in \mathcal{Q}S$, the set Q is closed, and therefore $Q' \in \mathcal{O}S$.

Moreover, since every T_2 space is strongly stacked, we have

$$v_\nabla = [Q'].$$

As $Q' \in \mathcal{O}S$, it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_\nabla} = (\mathcal{O}S)_{u_{Q'}} \cong\uparrow Q'.$$

Now, $u_{Q'}$ is a complemented nucleus in $\mathcal{O}S$, by [4, Proposition VI.3.3], implies that $\mathcal{O}S_{v_\nabla}$ is spatial. Therefore,

$$(\mathcal{O}S)_{v_\nabla} \cong \mathcal{O}\operatorname{pt}((\mathcal{O}S)_{v_\nabla}).$$

Since

$$\operatorname{pt}((\mathcal{O}S)_{v_\nabla}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_\nabla} \cong\uparrow Q'.$$

□

4. Interaction among Compact-Quotient Separation Properties

DONE: Ejemplo de neat no KC

DONE: Comprobar si neat implica stacked

DONE: KC implica S.S.?

Existe alguna propiedad P tal que (H)+P implica staked

T1+S.+P implica (H)?

DONE: Relacionar S. y S.S. con fit y sfit

We begin by recalling a relation established by Sexton in the spatial setting. By [6, Theorem 6.2 (a)], for every topological space S,

$$\mathcal{O}S \text{ neat} \implies S \text{ strongly stacked} \implies S \text{ stacked}.$$

The first of these implications provides a natural starting point for comparing neatness and strong stackedness in the point-free setting. Indeed, although neatness of a frame does not directly imply strong stackedness of the frame itself, it does imply strong stackedness of its point space.

Proposition 4.1. *If A is a neat frame, then $\operatorname{pt} A$ is strongly stacked.*

Proof. Let $S = \operatorname{pt} A$. Since neatness is hereditary, $\mathcal{O}S$ is neat. By [6, Theorem 6.2 (a)], S is strongly stacked. □

The preceding proposition concerns the point space of A, rather than the frame A itself. It is therefore natural to ask under what additional conditions neatness of A implies that A is strongly stacked. The comparison induced by the spatial reflection provides such a condition. For every HM triple (F, Q, G), we have

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Suppose now that A is strongly stacked. Then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since

$$v_G \leq [Q'],$$

we obtain

$$\mathcal{U}_* \circ v_G \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F.$$

Together with the previous inequality, this yields

$$v_F = \mathcal{U}_* \circ v_G \circ \mathcal{U},$$

or equivalently,

$$\mathcal{U} \circ v_F = v_G \circ \mathcal{U}.$$

Thus, strong stackedness forces the comparison induced by the spatial reflection to commute. Under neatness, the converse also holds.

Proposition 4.2. *Let A be a neat frame. If*

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$$

for every HM triple (F, Q, ∇), then A is strongly stacked.

Proof. Let $S = \operatorname{pt} A$ and let (F, Q, ∇) be an HM triple.

Since A is neat, $\mathcal{O}S$ is neat. Hence S is stacked and packed. In particular, every compact saturated subset $Q \subseteq S$ is closed, so

$$Q' \in \mathcal{O}S.$$

Since S is stacked,

$$Q = \partial_\nabla^\infty(S),$$

and therefore, by Lemma 3.4,

$$v_\nabla(\varnothing) = Q'.$$

Because $\mathcal{O}S$ is neat, it follows that

$$v_\nabla = u_{Q'} = [Q'].$$

By hypothesis,

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U} = [Q'] \circ \mathcal{U}.$$

Applying the right adjoint $\mathcal{U}_*$, we obtain

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore, A is strongly stacked. □

We next turn to the KC property and compare it with the preceding notions. We first observe that KC always implies neatness.

Proposition 4.3. *Let A be a KC-frame. Then A is neat.*

Proof. Let $F \in A^\wedge$ and put

$$d = v_F(0).$$

Since A is KC and

$$v_F \in [v_F, w_F],$$

there exists $a \in A$ such that

$$v_F = u_a.$$

Evaluating at 0, we obtain

$$d = v_F(0) = u_a(0) = a.$$

Hence

$$v_F = u_d.$$

Since this holds for every $F \in A^\wedge$, the frame A is neat. □

Proposition 4.4. *Let A be a KC frame. Then A is stacked if and only if it is strongly stacked.*

Proof. Let (F, Q) be an HM-pair and put

$$d = v_F(0) \qquad \text{and} \qquad a = w_F(0).$$

Since every KC-frame is T_1, we have

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F.$$

Moreover, by the previous proposition,

$$v_F = u_d.$$

Since A is KC, w_F is closed as well, and therefore

$$w_F = u_a.$$

Consequently,

$$\begin{aligned} A \text{ is strongly stacked} &\iff v_F = w_F \\ &\iff u_d = u_a \\ &\iff d = a \\ &\iff v_F(0) = w_F(0) \\ &\iff A \text{ is stacked.} \end{aligned}$$

Since this holds for every HM-pair (F, Q), the result follows. □

The preceding proposition does not assert that every KC-frame is stacked. In fact, the following example shows that KC does not imply even stackedness.

Example 4.5.

Proposition 4.6. *Let A be a spatial frame. Then A is KC if and only if it is neat.*

Proof. Suppose first that A is KC. By the previous results, every KC frame is neat.

Conversely, let A be neat. Since A is spatial, we may write

$$A \cong \mathcal{O}(S)$$

for some topological space S. By the spatial characterization of neatnees, S is packed and stacked. Moreover, neatness implies that S is strongly stacked. Hence S is packed and strongly stacked, and therefore $\mathcal{O}(S)$ is KC. Thus A is KC. □

The preceding results show that KC imposes strong restrictions on the relation between neatness and the two forms of stackedness. We now consider converses under additional separation hypotheses.

Proposition 4.7. *Let A be a KC frame. If A is subfit, then A is strongly stacked.*

Proof. Let (F, Q) be an HM-pair and put

$$d = v_F(0) \qquad \text{and} \qquad a = w_F(0).$$

Since A is KC, both v_F and w_F are closed nuclei. Hence

$$v_F = u_d \qquad \text{and} \qquad w_F = u_a.$$

By subfitness and [10, Theorem 3.5], we have

$$d = a.$$

Therefore,

$$v_F = u_d = u_a = w_F.$$

Since A is T_1, we also have

$$w_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Consequently,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since this holds for every HM pair (F, Q), A is strongly stacked. □

Proposition 4.8. *Let A be a frame. If A is neat and fit, then A is KC.*

Proof. Let $F \in A^\wedge$ and $j \in NA$ be such that

$$\nabla(j) = F.$$

Since A is fit, by [10, Theorem 3.4], every nucleus on A is fitted. Hence j is the least nucleus in its block. On the other hand, v_F is also the least nucleus admitting F. Therefore,

$$j = v_F.$$

Since A is neat, there exists $d \in A$ such that

$$v_F = u_d.$$

Consequently,

$$j = u_d.$$

Thus every nucleus whose associated filter is open is closed, and hence A is KC. □

Proposition 4.9. *Every neat strongly stacked frame is KC.*

Proof. Let $F \in A^\wedge$. Since A is neat, it is T_1. Hence, as A is strongly stacked, we have

$$v_F = w_F.$$

Moreover, of neatness,

$$v_F = u_d, \qquad d = v_F(0).$$

Therefore,

$$[v_F, w_F] = \{u_d\}.$$

Since every nucleus inducing a compact quotient of A belongs to an admissibility interval $[v_F, w_F]$ for some $F \in A^\wedge$, every such nucleus is closed. Hence every compact quotient of A is closed, and therefore A is KC. □

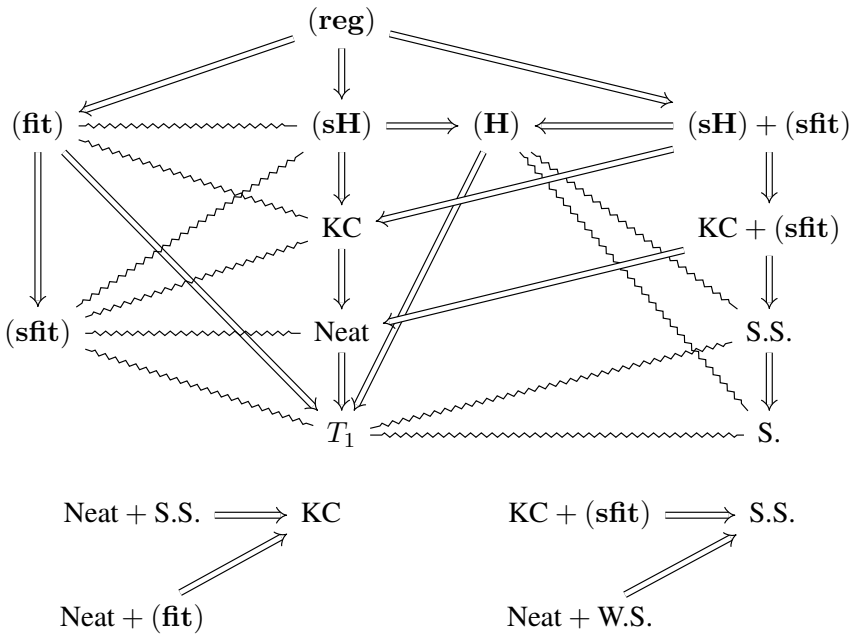
Proposition 4.6 shows that neatness and the KC property coincide for spatial frames. However, this equivalence does not extend to arbitrary frames. The following theorem provides a non-spatial neat frame admitting a compact quotient which is not closed, thereby establishing that the implication from KC to neatness is strict in the point-free setting.

Theorem 4.10. *There exists a non-spatial neat frame which is not KC. Consequently, the implication*

$$\mathrm{KC} \Longrightarrow \mathrm{Neat}$$

is strict in the category of frames.

Proof. □



5. Relationship with the patch assembly

Following the idea presented in [10] (or, alternatively, in [9]), let us consider the linearly ordered frame

$$A = [0, 1].$$

In this case the following observations hold.

(1) For all $a, b \in A$ the implication $(a \succ b)$ is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

(2) For $a, x \in A$, the distinguished nuclei u_a, v_a, w_a are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases} \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

(3) If $F \subseteq A$ is a filter, then F is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some $a \in A$. In particular, $F \in A^\wedge$ if and only if $F = (a, 1]$.

(4) If $F \in A^\wedge$, the adjusted nucleus associated v_F is calculated as

$$v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases} \tag{8}$$

where $l_m = v_m \wedge w_m$ and $m = \bigwedge F$.

(5) For $S = \operatorname{pt} A$ it holds that $S = [0, 1)$ and the open sets in $\mathcal{O}S$ are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

(6) The frame A is not T_1, since no $p \in S$ is maximal. Consequently, A does not satisfy any stronger separation property in Frm.

(7) For $F \in A^\wedge$ the compact saturated set associated $Q \in \mathcal{Q}S$ is of the form $Q = [0, m]$.

(8) It holds that $[Q'] = v_F$; therefore, S is strongly stacked.

(9) Since $[0, x) \in \mathcal{O}S$, then $[x, 1] \in \mathcal{C}S$ and $Q = [0, m]$. Therefore, the compact subsets are not closed; that is, S is not packed.

(10) The frame A is not efficient.

The above resume some characteristics about the frame $[0, 1]$. Now, we will perform the analysis of the patch constructions of A and its point space S.

Remember that for $A \in \mathrm{Frm}$ and $S \in \mathrm{Top}$ we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where Pbase generates the patch frame of A and pbase generates the topology of the patch space of S. By (8) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1)\}.$$

For $x \in A$ we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of x and m.

Case 1: If $x < m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

Case 2: If $x = m$, then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

Case 3: If $x > m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of a does not affect the evaluation. Thus, the only non-trivial case occurs when $m < x$, where the value of $(u_a \wedge l_m)(x)$ depends on the relative position of a and x. We therefore distinguish the following two subcases.

Case 3.1: If $m < x < a$, then

$$(u_a \wedge l_m)(x) = a.$$

Case 3.2: If $m < a \leq x$, then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of Pbase correspond precisely to the pairs (m, a) with $m < a$. Consequently, the frame PA is determined by the open intervals of the form (m, a), where $m \in \operatorname{pt} A$ and $a \in A$.

We now turn to the question of whether PA satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let A be an arbitrary frame and let $S = \operatorname{pt} A$ be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc}
A & \longrightarrow & PA & \hookrightarrow & NA \\
{\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle P(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\
\mathcal{O}S & \longrightarrow & P\mathcal{O}S & \hookrightarrow & N\mathcal{O}S \\
& \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\
& & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S
\end{array}$$

where $\mathcal{O}^f S$ is the Skula topology (or front topology) of S. In [7] is named *the patch assembly diagram* and more details about it can also be found in [**?**].

In our case, $\mathcal{U}_A$ is an isomorphism. Moreover, under $\mathcal{U}_A$, the patch construction turns out to be functorial and, therefore, $PA \cong P\mathcal{O}S$. Similarly, $NA \cong N\mathcal{O}S$. To describe explicitly $P\mathcal{O}S$ and $N\mathcal{O}S$, we use the following result, which can be consulted in [1].

Proposition 5.1. *Let S be a partially ordered set.*

(1) *If S has no infinite antichains, then $N\mathcal{O}S$ is spatial.*
(2) *If S is totally ordered, then $N\mathcal{O}S$ is spatial.*

Thus, by Proposition 5.1 we have that $N\mathcal{O}S$ is spatial and therefore $N\mathcal{O}S \cong \mathcal{O}^f S$. With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong\downarrow & & \downarrow\cong & & \downarrow\cong \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame PA, we can study the patch topology $\mathcal{O}^p S$. Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0,a) \cap [m,1) \mid a \in [0,1], m \in [0,1)\} = \{(m,a)\}.$$

Since the topology generated by the intervals (m,a) coincides with the usual topology on $[0,1)$, the space ${}^p S$ is Hausdorff. Consequently, $\mathcal{O}^p S \cong PA$ satisfies (**H**), even when A is not T_1 nor neat.

Lemma 5.2. *Let A be a frame, if the morphism $\mathrm{p}U\colon \mathrm{p}A \to \mathrm{p}\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q, thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we neee to observe the action of $\mathrm{p}U$ on these two nuclei, in order to do that remember that we have a conmutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & \mathrm{p}A & \xhookrightarrow{\iota} & NA \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle \mathrm{p}(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\ \mathcal{O}S & \longrightarrow & \mathrm{p}\mathcal{O}S & \xhookrightarrow[\imath]{} & N\mathcal{O}S \\ & \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

since $N(\mathcal{U})\iota = \imath \mathrm{p}\mathcal{U}$, that is, $\mathrm{p}\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to $\mathrm{p}A$, then

$$\mathrm{p}\mathcal{U}(\mathrm{v}_F) = N(\mathcal{U})(\mathrm{v}_F) = \bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee\{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $\mathrm{v}_{\mathcal{U}F}$. □

Completar esta prueba

Considerar incluir información sobre los parches

REFERENCES

1. Francisco Ávila, Guram Bezhanishvili, PJ Morandi, and Angel Zaldívar, *The frame of nuclei on an alexandorff space*, Order (2020).
2. Melvin Hochster, *Prime ideal structure in commutative rings*, Transactions of the American Mathematical Society **142** (1969), 43–60.
3. Jan Paseka and Bohumil Šmarda, *T_2-frames and almost compact frames*, Czechoslovak Mathematical Journal **42** (1992), no. 3, 385–402.
4. J. Picado and A. Pultr, *Frames and locales: Topology without points*, Frontiers in Mathematics, Springer Basel, 2012.
5. Jorge Picado and Aleš Pultr, *Separation in point-free topology*, Springer, 2021.
6. R. A. Sexton and H. Simmons, *An ordinal indexed hierarchy of separation properties*, Topology Proceedings **30** (2006), no. 2, 585–625.
7. RA Sexton and H Simmons, *Point-sensitive and point-free patch constructions*, Journal of Pure and Applied Algebra **207** (2006), no. 2, 433–468.
8. Rosemary A. Sexton, *A point-free and point-sensitive analysis of the patch assembly*, Ph.D. thesis, The Victoria University of Manchester, 2003.
9. Harold Simmons, *The vietoris modifications of a frame*, Unpublished manuscript, 79pp., available online at http://www. cs. man. ac. uk/hsimmons (2004).
10. _____, *Regularity, fitness, and the block structure of frames*, Applied Categorical Structures **14** (2006), 1–34.
11. Albert Wilansky, *Between T_1 and T_2*, The American Mathematical Monthly **74** (1967), no. 3, 261–266.